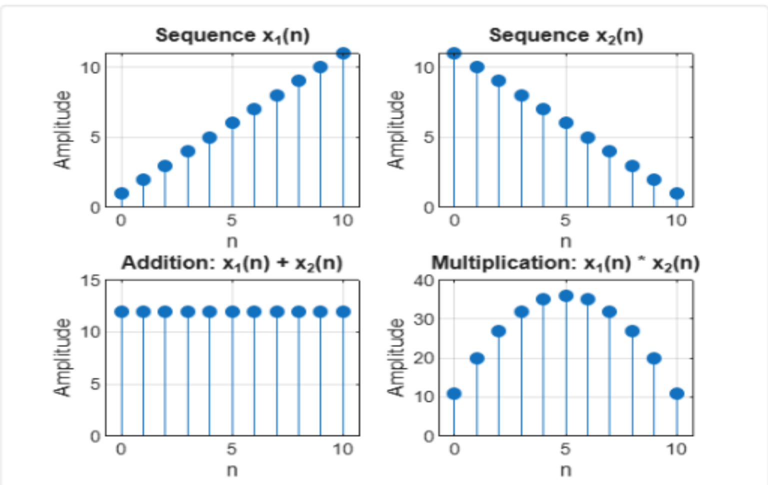
3. SEQUENCE OPERATIONS AND SIGNAL ANALYSIS

Experiment 3.1

```
% Addition and Multiplication of Two Sequences
clc;
clear;
close all;
% Sample index
n = 0:10;
% Sequence 1

x1 = [1 2 3 4 5 6 7 8 9 10 11];
% Sequence 2
x2 = [11 10 9 8 7 6 5 4 3 2 1];
% Addition of sequences
y_add = x1 + x2;
% Multiplication of sequences
y_mul = x1 .* x2;
% Plotting
figure;
subplot(2,2,1);
stem(n,x1,'filled');
grid on;
title('Sequence x_1(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,2);
```

```
stem(n,x2,'filled');
grid on;
title('Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,3);
stem(n,y_add,'filled');
grid on;
title('Addition: x_1(n) + x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,4);
stem(n,y_mul,'filled');
grid on;
title('Multiplication: x_1(n) * x_2(n)');
xlabel('n');
ylabel('Amplitude');
```



Experiment 3.2

```
% Energy and Power Calculation of Signals
clc;
clear;
close all;

% Time index
n = -15:15;     % 31 samples

% Signal
x = [zeros(1,10) 1 2 3 4 5 4 3 2 1 zeros(1,10)];

% Energy Calculation
E = sum(abs(x).^2);

% Power Calculation
P = E / length(x);

% Display Results
disp(['Energy of the signal = ', num2str(E)]);
disp(['Power of the signal = ', num2str(P)]);

% Plot Signal
```

```
figure;
stem(n, x, 'filled');
grid on;
title('Discrete-Time Signal');
xlabel('n');
ylabel('Amplitude');
```

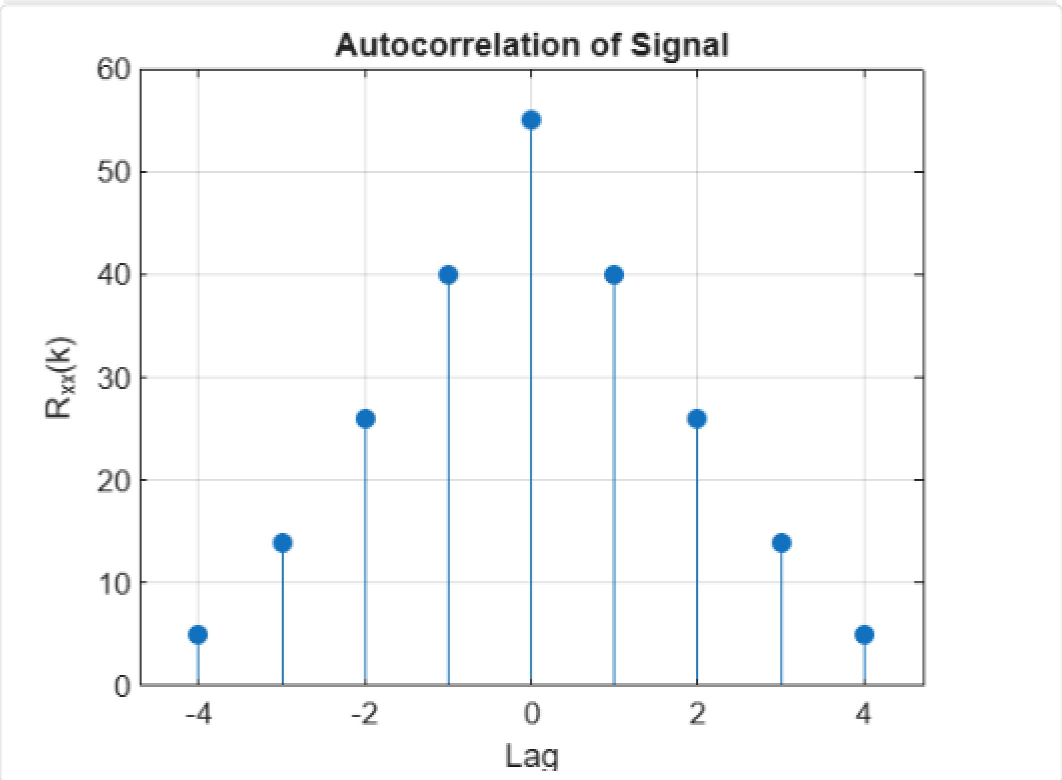
```
Energy of the signal = 85
Power of the signal = 2.931
```

Experiment 3.3

```
% Autocorrelation of a Signal
clc;
clear;
close all;
% Input Signal
x = [1 2 3 4 5];
% Autocorrelation
Rxx = xcorr(x);
% Lag values
lags = -(length(x)-1) :(length(x)-1);
% Display result
disp('Autocorrelation Sequence:');
```

```
disp(Rxx);

% Plot

figure;

stem(lags,Rxx,'filled');

grid on;

title('Autocorrelation of Signal');

xlabel('Lag');

ylabel('R_{xx}(k)');
```

```
Autocorrelation Sequence:
     5.0000   14.0000   26.0000   40.0000   55.0000   40.0000   26.0000
```



Experiment 3.4

```
% Cross Correlation of Two Signals
clc;
clear;
close all;
% Input Signals
x = [1 2 3 4 5];
y = [5 4 3 2 1];
% Cross Correlation
Rxy = xcorr(x,y);
% Lag values
lags = -(length(x)-1):(length(x)-1);
% Display result
disp('Cross Correlation Sequence:');
disp(Rxy);
% Plot
figure;
stem(lags,Rxy,'filled');
grid on;
title('Cross Correlation of Two Signals');
xlabel('Lag');
ylabel('R_{xy}(k)');
```

```
Cross Correlation Sequence:
     1.0000      4.0000     10.0000     20.0000     35.0000     44.0000     46.0(
```